

Cheeger Bottleneck Priority: Method Note

Cheeger bottleneck priority

Plain-language purpose

This layer marks places where the modeled urban heat-and-cooling network is weakly connected. It is a **prioritization signal**, not a diagnosis of any person, block, or neighborhood.

Inputs

- A land-surface-temperature raster, normalized to a 0–1 scale.
- An optional vegetation index (NDVI), also normalized to 0–1.
- Valid raster cells, joined to their four or eight neighboring cells.

Step 1 – build a weighted graph

Every valid raster cell becomes a node. Neighboring cells are joined by an edge with conductance

$$w_{ij} = \exp(-\alpha \times g_{ij}) \times (1 + \beta \times \text{NDVI}_{ij}) .$$

Here g_{ij} is the mean local temperature-gradient magnitude across the two cells. Sharp heat changes lower conductance; greener connections can raise it. The published pipeline defaults are

$$\alpha = 3.0 \text{ and } \beta = 0.6 .$$

Step 2 – find the mathematically weak seam

Let A be the weighted adjacency matrix and D the diagonal degree matrix. The normalized graph Laplacian is

$$L_{\text{norm}} = D^{(-1/2)} (D - A) D^{(-1/2)} .$$

We compute its second-smallest eigenvalue and corresponding Fiedler vector. Sorting cells by that vector produces candidate partitions. For each candidate set S , the pipeline calculates

conductance

$$\text{phi}(S) = \text{cut}(S, S^c) / \min(\text{vol}(S), \text{vol}(S^c)) .$$

The lowest-conductance sweep cut is the Cheeger-style bottleneck. The displayed cells are the boundary of that cut, expanded slightly for a readable geographic output.

Step 3 – turn a seam into a priority

Only cells on the bottleneck boundary receive a priority score:

$$\text{priority} = 0.65 \times \text{normalized_heat} + 0.35 \times (1 - \text{cooling_access}) .$$

The value is clipped to `[0, 1]` and exported on a 0-100 display scale. The score says **where to investigate first**, not that an intervention will produce a particular temperature reduction.

How to interpret responsibly

- Compare with local shade, housing, health, access, and community knowledge before acting.
- A low-conductance result can reflect raster resolution, missing cells, or parameter choices.
- The result is not a causal estimate, exposure measurement, or individual-level risk score.

Reproducibility

Implementation: `core/graph.py` , `core/spectra.py` , and `core/pipeline.py` . Run configuration, input dates, raster resolution, missing-data treatment, and parameters must accompany any public claim.